

cell0_recip_stack to Authoritative cell3 Bridge

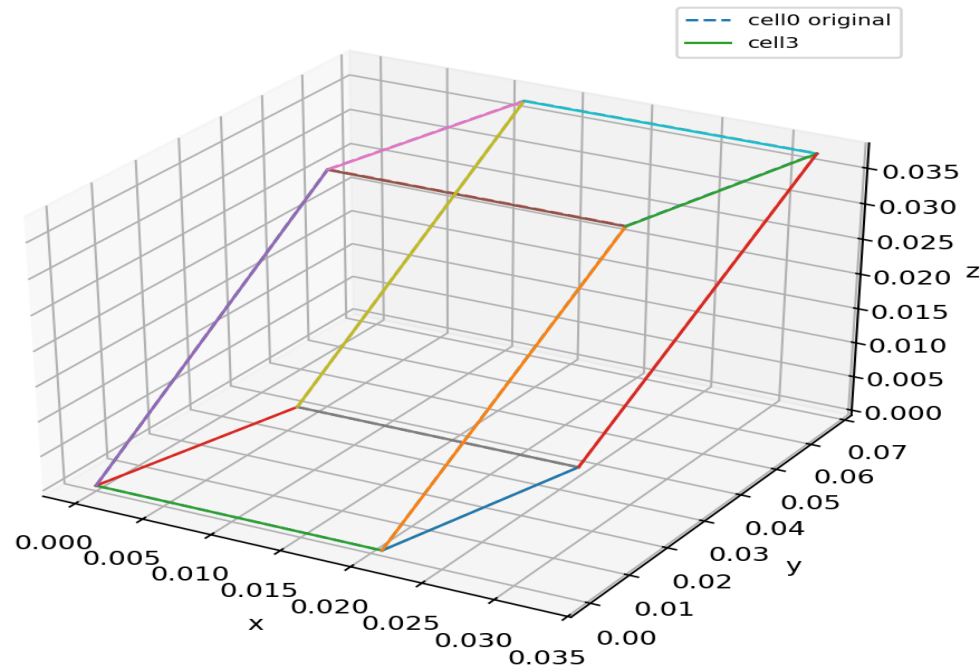
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Bridge provenance

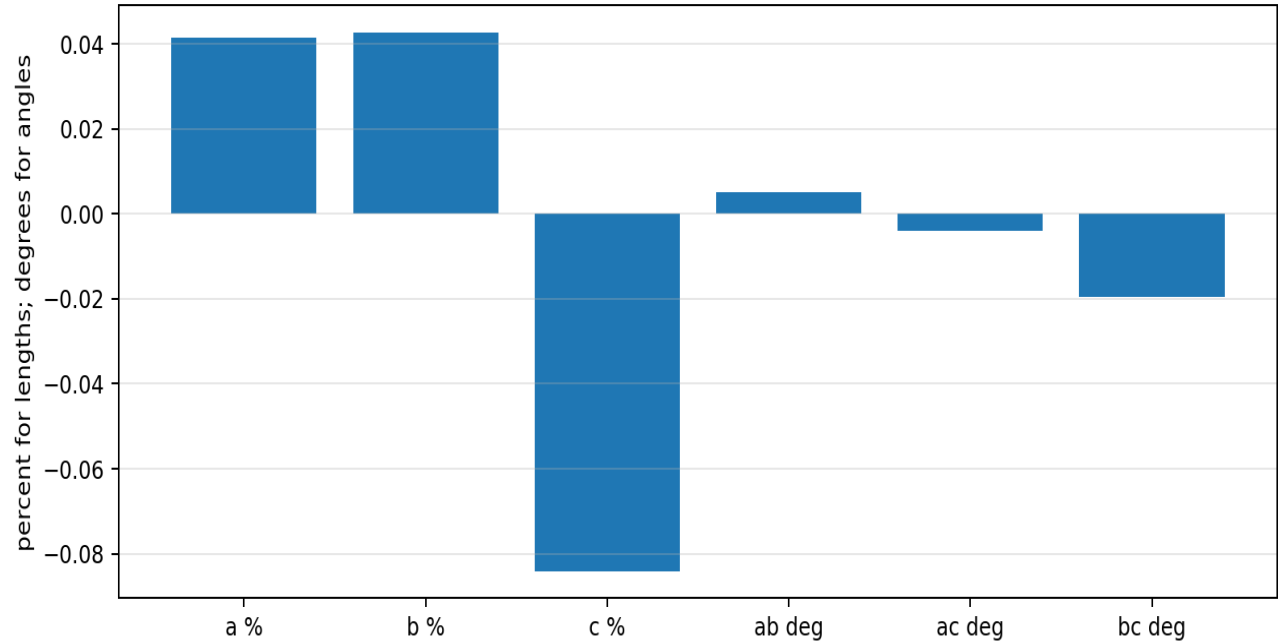
This paper begins after the native galaxy cell0 derivation is complete. It measures the perturbation needed to move from cell0/original geometry to authoritative cell3. This is not a substitution step; it is a controlled perturbation bridge.

Cell0 and cell3 geometry comparison

Canonical perturbation overlay: cell0 to cell3



Perturbation from native cell0 to authoritative cell3



quantity	cell0	cell3	delta cell3-cell0	relative delta
a	0.020788448186	0.020797058782	8.610595177719e-06	4.142009591336e-04
b	0.036478622795	0.036494205131	1.558233586020e-05	4.271634910091e-04
c	0.0493500623	0.0493085659	-4.149639956731e-05	-8.408580989254e-04
theta_ab	83.928361201001	83.933483012313	0.005121811313	6.102598977779e-05
theta_ac	80.452247903308	80.448130018338	-0.00411788497	-5.118421271327e-05
theta_bc	50.382886569535	50.363231813756	-0.019654755779	-3.901077750271e-04
A_ab	7.540800268680e-04	7.547217896399e-04	6.417627718418e-07	8.510539319112e-04
A_ac	0.001011699986	0.001011255758	-4.442288152344e-07	-4.390914511954e-04
A_bc	0.001386752323	0.001385784432	-9.678909767220e-07	-6.979551869982e-04
volume	2.842908885472e-05	2.842066937711e-05	-8.419477606914e-09	-2.961571385541e-04
surface_area	0.006305064672	0.006303523958	-1.540714040228e-06	-2.443613381228e-04

Native bridge perturbation

bridge	cell0	cell3 target	delta target-cell0
A_bc/A_ac	1.37071497604	1.37035999177	-3.549842699773e-04
A_bc/A_ab	1.838998877184	1.836152673426	-0.002846203758
A_ac/A_ab	1.341634774063	1.339905341993	-0.00172943207
eta_N	1.838998877184e+03	1.836152673426e+03	-2.846203757523
c_N	137.071497603998	137.035999177	-0.035498426998
alpha_N	0.007295462715	0.007297352564	1.889849763148e-06

Interpretation

The perturbation scale is consistent with the uploaded cell2/cell3 provenance chain. The native galaxy-derived cell0 is the original/Lattice2-like cell. The bridge from cell0 to authoritative cell3 passes through the documented refinement path: original cell0/Lattice2 -> cell2 small local deformation -> cell3 ppb-scale final correction. Thus the student first derives the native galaxy cell and then learns why a controlled perturbation is needed to reach the authoritative closure geometry.

Audit

Gate	Status
cell0 geometry is reported before perturbation	PASS
cell3 appears only as bridge target	PASS
all deltas are computed directly from cell0 and cell3 tables	PASS
perturbations are small and consistent with Lattice2 -> cell2 -> cell3 provenance	PASS
ReportLab core fonts only	PASS